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A Systematic Framework for Measuring Team Performance and Group Decision-Making: Pitfalls, Solutions, and Transdisciplinary Insights

Ignacio VAZQUEZ[a,1], Hüseyin AKDOGAN[b], Marc-Olivier DEL GROSSO[b], Natalia ZWARTS[b]
[a] *Massachusetts Institute of Technology, USA*
[b] *The Hague University of Applied Sciences, The Netherlands*

Abstract. Evaluating and enhancing team performance is a cornerstone of transdisciplinary engineering, yet the absence of a formal method to categorize and assess the efficacy of group performance experiments remains a significant challenge. In particular, what kinds of results can an experiment on team performance produce, and how can researchers ensure these results remain epistemologically sound? This study addresses these foundational questions by proposing a structured framework to categorize and assess team performance experiments. The framework integrates cognitive measurements, statistical rigor, and meta-analytical evaluations of experimental maturity to enhance the credibility and reliability of study outcomes. The approach is illustrated through the analysis of data from two team performance experiments, serving two primary objectives. First, the analysis demonstrates the actionable insights into team dynamics enabled by the framework, showcasing its practical applicability. Second, it systematically addresses critical factors influencing experimental reliability, including the identification of methodological pitfalls, thereby offering tools and strategies to enhance the design and evaluation of team performance studies. This work bridges the gap between theoretical concepts and practical implementation, contributing to the advancement of methodologies for transdisciplinary teamwork and performance measurement. The findings are particularly relevant to researchers and practitioners seeking robust, reproducible methodologies for assessing team dynamics across diverse disciplines, offering a guideline for future studies aimed at refining team performance evaluation methods.

Keywords. Team performance, group decision making, experiment framework.

Introduction

Methodologies used to design experiments on team performance focus on different aspects of evaluations. Despite the collective nature of teamwork, individual attributes are increasingly investigated to determine their effects on team performance [1]. The environment where the measurement is done is also evolving with the changes of the workplace. Due to the digitalization of work and education, researchers are exploring the characteristics of virtual teams to aid problem-solving [2]. Today's average college

[1] Corresponding Author, Email: ignaciov@mit.edu.

graduates have spent less than 5,000 h of their lives reading, but over 10,000 h playing video games [3]. Learning methods are catching up with the change of lifestyle, introducing websites, apps, chats and simulations that stimulate group performance. This in turn generates groups with higher level of experience in using new digital tools, potentially creating a trade-off between task repetition and adaptation [4].

The existing simulations have been broadly investigated to establish their effectiveness, looking at both group results and team dynamics [5]. Team dynamics are also analysed in relation to team composition, noting the influence of gender [6], time zones [7], neurodivergence [8] and previous experience [9]. Researchers also experimented with factors such as negative mood, introducing contextual measurements into studies [10].

Next to different perspectives on measured data points, there are a number of additional variables which can be included in the assessment: time spent in the simulation, number of attempts in completing tests, individual contributions as opposed to group outcomes, retention rate for learning [11]. Further experiments have shown that effectiveness of simulations increases if a facilitator was guiding the group, and that observation has similar effects as active participation [12]. This leads to questions regarding design of the experiment, selection of the methodology and considerations to refine the evaluation of the results.

Despite a multitude of scientific experiments and practical developments, there is no formal method to categorize and evaluate the methodology of executing group performance experiments. This paper aims at providing conclusions from four disciplines relevant to credibility of the experiment; System Design, Cognitive Measurements, Statistical Analysis, and Meta-analysis of Experiments Maturity levels.

The conclusions are relevant to validation of the results and internal validity of the future studies. In the past, a number of concerns have been raised in relation to the scale of the experiment, data reliability, confounding factors, biases in data collection and performance definition. Most notably, two experiments addressing the same question could yield opposite results. In the case of flexdesk investigations comparing communication, satisfaction and performance, the first study found exclusively positive effects of the introduction of open spaces [13]. A second, largely more reviewed experiment, has established diminishing personal communication, decline in productivity and well-being of employees [14].

The measurements proposed in the first study were declarative, so employees opted to assess their own communication and performance as high as possible. In the second study, a device was measuring the volume and format of communication. It also introduced Key Performance Indicators, objectively relating the output to previous baseline. The experimental design has proven to bring more truthful results when composed of less biased measurements.

This study presents a novel, systematic framework for evaluating team performance experiments that integrates cognitive science, statistical analysis, system design, and meta-analytic techniques. Unlike prior research that often addresses team dynamics through siloed or discipline-specific lenses, this paper offers a transdisciplinary method that identifies methodological pitfalls and proposes practical solutions to enhance the credibility and reproducibility of findings. A key contribution is the dual-layered meta-analysis of 49 experiments using a Mars colony simulation, which exposed high heterogeneity in team performance results—an outcome the authors attribute to inconsistent experimental designs and varied data collection practices. The framework's application not only reveals the significance of structured emotional and cognitive feedback tools but also provides guidelines for designing more robust experiments. This positions the study as a critical stepping stone toward standardizing team performance evaluation across diverse settings and disciplines.

1. System Design and Experiments on Performance

The experiment relies on a role-playing game where participants become stakeholders in a team aimed at finalizing the design of the first Mars Colony [48]. The motivation of having said scenario is to create; (1) a standard set of conditions for all the teams that suspend the reality outside of the game and hampers the influence of specific disciplinary experience and given individual participant may have, (2) provide the same starting point in terms of information and (3) create a basic expectation of how to use the system model given to the team, represented in the experiment by the Mars simulation.

The Mars colony scenario sets up a shared goal within a fictional organization. The team's work is framed as being part of a company-wide mission run by the world's largest (fictional) aerospace company and setting their team goal to choose a site design for Star City's first settlement. Primary Team Mission: Find a design that will lead to a successful Mars Colony. This setup allows for an in-depth analysis of the guiding research question (RQ) for the experiments: To what extent do cognitive factors like Theory of Mind (ToM) and demographic diversity affect team performance outcomes in simulation-based experimental settings?

One of the purposes of using a simulation model is to eliminate the subjectivity of team performance. It is not that being subjective is wrong, but the goal here is to reduce the noise so that we can isolate the contribution of team communication, emotion, and their influence over the system performance. The problem was designed so that it can be easily understood by general non-engineering participants in a short time —about five minutes. The problem description was provided in a single-pager and a facilitator briefing. The instructions to use the system model were provided as a video recording.

The game played by the participants is an interpretation of [49] adapted to team-level teamwork studies.

2. Cognitive Measurements

A key topic in cognitive science is how groups form and update a shared mental model, which guides cooperative decision-making [15] [16].

The study's strength lies in its emphasis on real-time team interactions and social-cognitive processes. The Mars simulation enables a direct observation of how teams converge on a shared understanding of a problem. It also highlights the interplay between social cognition, collaborative problem-solving, and technological mediation (i.e., using real-time feedback tools).

In these experiments, it is possible to measure the extent to which teams can synchronize their strategies under time pressure and resource constraints, which is a fundamental aspect of real-life cooperative decision-making [17]. Analysing whether subgroups (e.g., mixed-gender teams or teams with higher average RME scores) perform differently can also give insights into Theory of Mind (ToM) skills and Group dynamics.

ToM is defined as the ability to perceive or infer others' mental states—and is proxied in our simulation by higher RME (Reading the Mind in the Eyes) score. It can facilitate faster alignment of mental models within a group. If teammates more accurately read each other's emotional or cognitive cues, they may need fewer explicit clarifications, thus reducing coordination overhead [18] [19]. Such ToM-related competencies help streamline communication, as fewer misunderstandings emerge regarding intentions and task divisions [20].

Mixed-demography teams sometimes display distinct patterns of communication or problem framing. This observation resonates with prior research showing that diverse teams can integrate a wider range of perspectives, though the empirical outcomes often depend on the task domain and the cultural context in which collaboration occurs [21] [22]. In some cases, gender-diverse groups have been found to engage in more balanced participation and broader information exchange [23], potentially enhancing collective decision-making when coupled with structured coordination strategies.

While these insights offer fresh contributions to our understanding of collaborative team dynamics, they also draw attention to potential confounding variables. It is thus necessary to place these findings into methodological perspective, examining how the design choices and measurement protocols either reinforce or undermine the robustness and generalizability of the observed effects.

3. Internal Validity Considerations

Internal validity reflects the degree to which observed effects can be attributed to the experimental manipulations rather than extraneous factors [24]. In this set of experiments—spanning multiple years (2022–2024) and locations—the cognitive/psychosocial measurements included real-time emotional feedback (using facial recognition software), post-experiment surveys, and, in some cases, the Reading the Mind in the Eyes (RME) test [25] [26]. Random assignment to control and treatment groups, as well as the uniformity of the central collaborative task, bolsters internal validity [27]. Nevertheless, uneven application of these cognitive measures across time and sites introduces potential confounds [28].

A first important consideration touches upon the consistency of treatment administration. Even minor variations in how participants receive instructions can influence engagement [29] [30]. If experimenters in one site emphasize the importance of emotional cues more than in another, participants' use of (and reliance on) the feedback can diverge [31]. We also realized, having teams in various countries, that post-experiment questionnaires are vulnerable to social desirability bias and cultural variations in reporting emotions [32] [33]. Differences in framing or language nuances across locations (e.g., USA's University 1 vs. The Netherlands' University 2) can yield systematic discrepancies in self-reported responses [34]. More broadly, it raises the issue of potential priming effects. For instance, all participants were shown a similar presentation video before engaging in the Mars simulation. Although the video was intended to introduce the task uniformly, it may have inadvertently primed participants to focus on specific aspects of teamwork or emotional states [35] [36]. With a different video mentioning collaboration strategies or the importance of emotion tracking, participants might have entered the simulation with heightened attention to these cues—thus reducing the natural variation in how they would otherwise approach problem-solving [37].

A second noteworthy element is the large between-study heterogeneity in meta-analyses [38]. This indicates that unstandardized measurement procedures can inflate variation in effect sizes [39]. For instance, small discrepancies in how the emotional feedback tools were presented (like phrasing the instructions differently) may partially explain high I^2 values observed in the “All Experiments” vs. “Selected Experiments” comparisons.

4. External Validity Considerations

External validity concerns whether findings apply beyond the specific context in which the study was conducted [40]. By spanning four institutions and employing participants with different cultural backgrounds, the dataset suggests a potential for broad applicability. However, academic settings with relatively young, high-performing individuals often differ significantly from general populations, echoing the critique of WEIRD (Western, Educated, Industrialized, Rich, and Democratic) samples [41]. Moreover, participation in these experimental studies was voluntary, which can lead to sampling bias [42]. Cultural norms regarding emotional displays can also affect how participants interact with, and interpret, real-time feedback [43].

An even more pressing question for our research team is how well these findings transfer to real-world scenarios, which often unfold over longer timescales and involve higher stakes than a controlled experimental setting. This is a necessary trade-off as the use of controlled “toy problems” is a staple in lab-based research for maximizing internal validity [24]. Yet, real-world team tasks often extend over months or years, include broader performance metrics, and involve high uncertainty—conditions that might dilute or alter the effect of brief interventions like real-time emotional feedback [44].

Lastly, the finding that mixed-gender teams align with prior evidence that diversity in perspectives can enhance team dynamics [45]. However, whether these effects hold in non-academic settings, culturally distinct organizations, or virtual teams operating across multiple time zones [46] remains an open question. High heterogeneity in the meta-analysis [38] may also indicate that cultural and temporal factors—and not just individual differences in social cognition—moderate the link between emotional feedback interventions and performance outcomes [39].

5. Methodology

This paper analyses two datasets collected through similar experimental designs. The first dataset, referred to as "All Experiments," includes 49 experiments conducted over three years in various locations.

1. The first set of experiments was conducted in 2022 at University 1 in the USA. These were exclusively in-person experiments.
2. The second set of experiments took place in 2023 at the University 2 and University 3 in the Netherlands. These were also conducted in person.
3. The third set of experiments, conducted in 2024, included locations at University 1, the University 2, University 3, and University 4 in Mexico.

The second dataset, referred to as "Selected Experiments," comprises experiments conducted exclusively in 2024 across the four universities.

The rationale for conducting two separate meta-analyses lies in the scope of the data collected across different years. Experiments conducted in 2022 and 2023 include only three descriptive variables, which are suitable for subgroup analysis; Type of Experiment, Year of Experiment and Location of Experiment. In contrast, the 2024 experiments incorporated additional data collection, including a post-experiment survey and RME (Reading the Mind in the Eyes) test results to assess the impact of neurodivergent team members on team dynamics. The additional descriptive variables in the 2024 dataset include:

- **Gender Mix**. Indicates whether the team consists of only one gender or is mixed-gender.
- **Group**. Differentiates between control and treatment groups.
- **Neurodivergent Composition**. Reflects the presence of neurodivergent members in the team based on RME test scores.
- **Experiment Participation**. Captures whether team members have prior experience in similar experiments.

The findings of the meta-analyses will now be presented, starting with the results for "All Experiments" (N=49), followed by the findings for "Selected Experiments" (N=26).

6. Results

The overall effect size for the meta-analysis was calculated under both fixed and random effects models. Table 1 summarizes the effect sizes, confidence intervals, Z-scores, and p-values for each model. The meta-analysis included a total of 49 experiments. Under the fixed effects model, the overall effect size was estimated at 0.213 with a 95% confidence interval (CI) ranging from 0.209 to 0.216. This effect was statistically significant with a Z-score of 130.50 and a corresponding p-value < 0.001, indicating that the observed effect size is unlikely to be due to chance.

For the random effects model, the effect size was near zero (0.0001), with a 95% CI from -0.283 to 0.283. The Z-score was 0.0006, and the p-value was 0.999, suggesting no significant effect. This outcome, combined with the high heterogeneity, highlights the variability across studies and emphasizes the need to explore potential moderators.

Table 1. Effect Size Estimate.

Model	k	Effect Size	95% CI (Low)	95% CI (High)	Z	p
Fixed	49	0.2126	0.2094	0.2157	130.4980	0.0000
Random	49	0.0001	-0.2828	0.2830	0.0006	0.9995

Figure 1 (left) presents a forest plot summarizing the effect sizes for each study, along with their 95% confidence intervals. Each horizontal line represents the CI for a specific study, with the central square indicating the study's point estimate. In this context, a negative slope represents an improvement in a team's Pareto Rank over time, suggesting better performance in terms of optimizing energy usage and resource allocation in the simulated Mars colony. Studies with effect sizes to the left of the zero line indicate improvements, while those to the right suggest no improvement or potential decline in performance.

Most individual studies exhibit effect sizes close to zero, with many confidence intervals intersecting the zero line. The substantial spread across studies visually reinforces the high heterogeneity identified statistically. The overall effect estimate, represented by a diamond at the bottom of the plot, aligns with the random effects model, suggesting a lack of significant effect due to high variability.

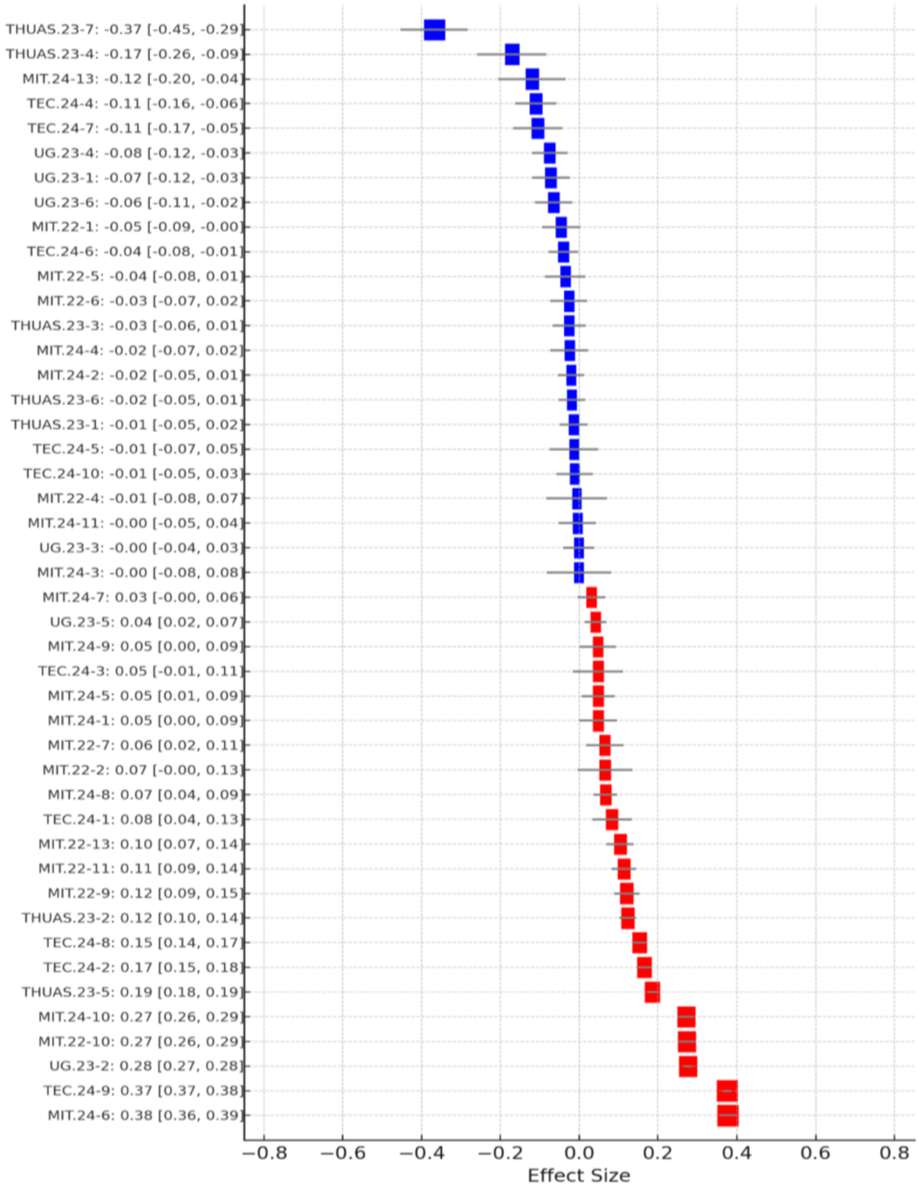

Figure 1. Forest plot providing a visual representation of the effect sizes for each study included in the meta-analysis, categorized by control and treatment groups across various subgroups (Location, Time, and Gender Mix). Each horizontal line represents the 95% confidence interval (CI) for the effect size of a particular study, with the midpoint square indicating the study's point estimate.

In this context, a negative effect size represents an improvement in a team's Pareto Rank over time, suggesting better performance in terms of optimizing energy usage and resource allocation in the simulated Mars colony. Therefore, studies with effect sizes to the left of the vertical zero line indicate teams that demonstrated improvement, whereas studies to the right suggest no improvement or potential decline in performance.

The forest plot reveals that most individual studies show effect sizes near zero, with confidence intervals that often intersect the no-effect line (the vertical line at zero), indicating variability in performance improvement. Studies with effect sizes to the left of the zero line (e.g., TBC-10, THUS-1) represent teams that demonstrated improvement in resource optimization and energy efficiency over time, with TBC-10 showing a relatively strong improvement effect at ES=−0.27. Conversely, studies to the right of the zero line suggest teams that either showed no improvement or potentially experienced a decline in performance.

However, high heterogeneity (I2=98.86%) indicates that this average effect does not fully capture the variability observed across individual studies. This high level of heterogeneity underscores the importance of exploring potential moderators and conducting subgroup analyses to understand the underlying factors driving differences in effect sizes.

In summary, the forest plot illustrates a general trend of improvement in teams' performance, particularly among certain subgroups. However, the wide confidence intervals for many studies and high heterogeneity suggest substantial variability, warranting further exploration of subgroup effects and potential moderators.

7. Conclusions

The findings both complement and challenge existing literature on team dynamics and performance. For instance, the study echoes the work of [1] and [18] by emphasizing the importance of shared mental models and temporal dynamics in team effectiveness. However, it extends this by directly incorporating ToM-based measures (e.g., RME scores) to operationalize social cognition during collaboration—something previous literature has often suggested but not methodically embedded into experimental designs. The observation that gender-diverse teams sometimes outperformed homogenous groups aligns with [23] findings on gender balance enhancing collective intelligence, yet the high heterogeneity in outcomes underscores [22] caution about context-dependency in diversity effects. The study also reveals how simulation fidelity and facilitator roles influence group behaviour, building on earlier insights from [3] about simulation effectiveness. Perhaps most notably, the stark contrast in fixed vs. random effect meta-analytic results mirrors the methodological debates raised by [14] on experimental consistency, highlighting the critical need for standardized, unbiased measurement protocols in group performance research.

A finding of this research is the high variability in observed outcomes, as demonstrated by the significant heterogeneity across experiments. While some studies indicate improvement in teamwork and problem-solving, other reveal inconsistencies due to methodological differences, cultural factors, and measurement biases. The fixed-effect model suggested s small but statistically significant effect, while the random-effects model showed no conclusive impact. This emphasized the need for more standardized experimental procedures.

The Mars Colony simulation provides a valuable baseline game for minimizing external influences and isolating the impact of team dynamics on performance by using a complex yet reachable model for non-experts. Nonetheless, variations in cognitive measurement techniques, such as real-time emotional feedback and RME test scores, suggest potential biases introduced by differing implementation methods across institutions and years. Internal validity considerations highlight challenges related to treatment administration consistency, social desirability bias, and the priming effects of instructional materials. Similarly, external validity concerns arose from the use of a constrained controlled experiment setting, which may not be representative of real-world team performance environments.

The findings underscore the necessity for refining experimental methodologies in future research. Standardizing measurement tools, expanding participant demographics, and incorporating long-term observational studies can enhance the generalizability and applicability of results. Additionally, leveraging meta-analytical techniques to identify moderating factor will help disentangle conflicting outcomes.

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